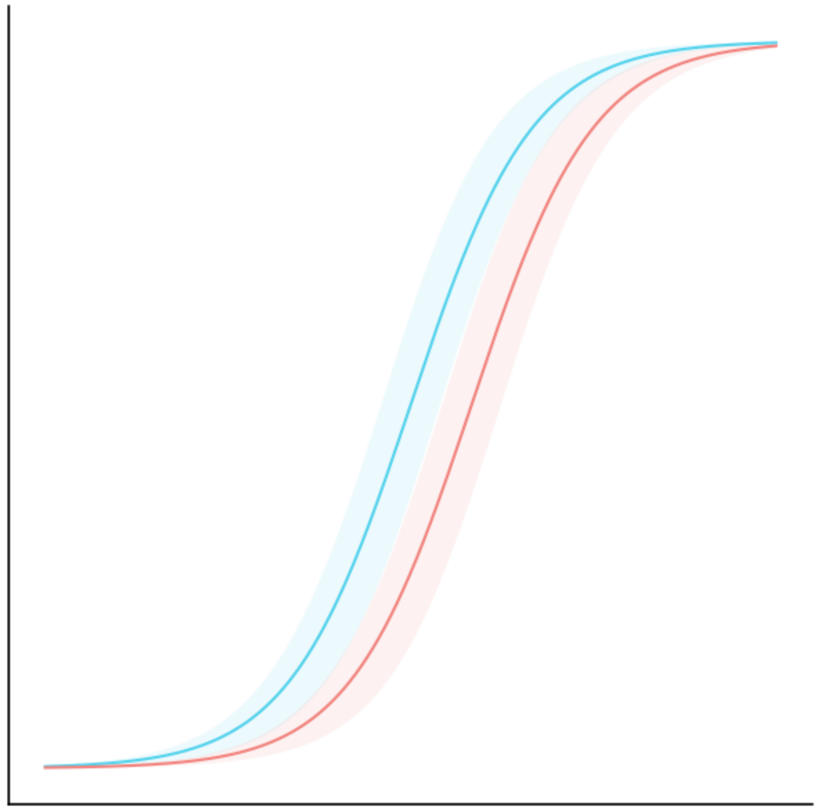
Logistic Regression Inference



Logistic Regression Inference - Example 1

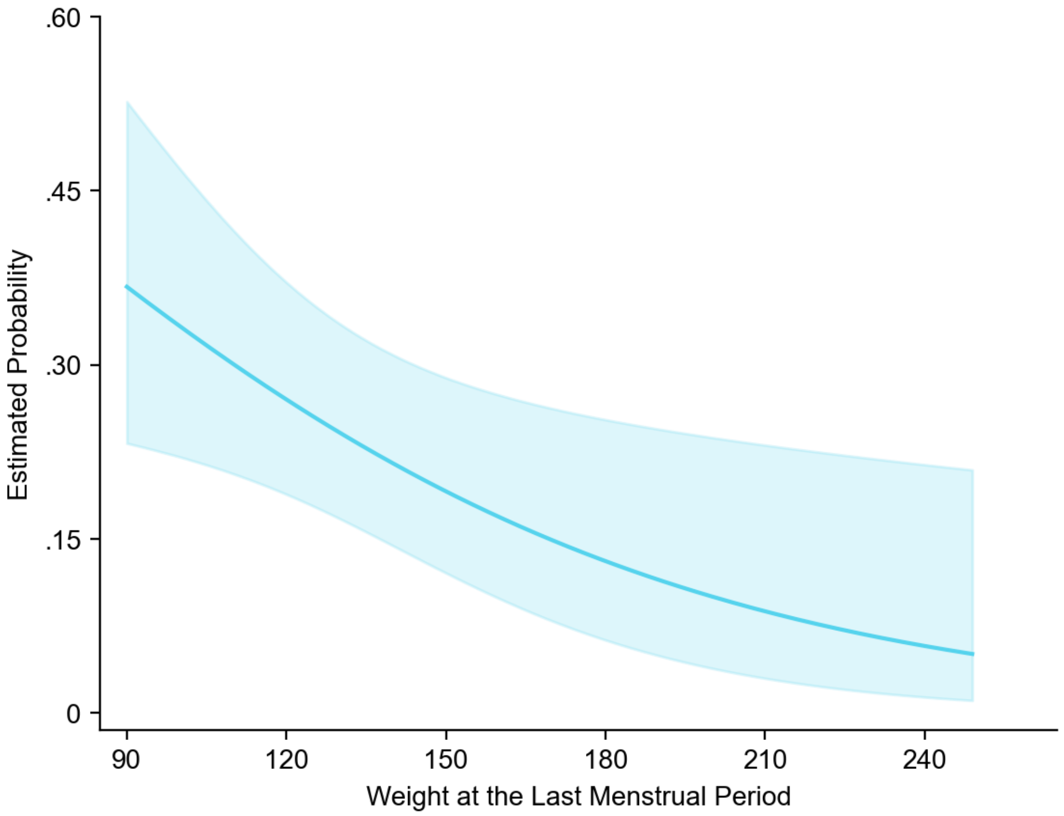
With binary logistic regression we can make inference about coefficients, predictions, and the `mean` response (e.g., how predictions and confidence intervals change depending on the inputs)

In Applied Logistic Regression by Hosmer, Lemeshow, and Sturdivant, a good example of the mean response estimation was provided for the Low Birth Weight Study

This analysis can be performed with statsmodels in Python, but requires a manual setup

Using `predict_ci` method of the logistic regression from `fisher-scoring`, we reconstruct the intervals as in the book (see the step-by-step example in the notebook)

Figure 3.5 Graph of the estimated probability of low weight birth and 95 percent confidence intervals as a function of weight at the last menstrual period for white women.



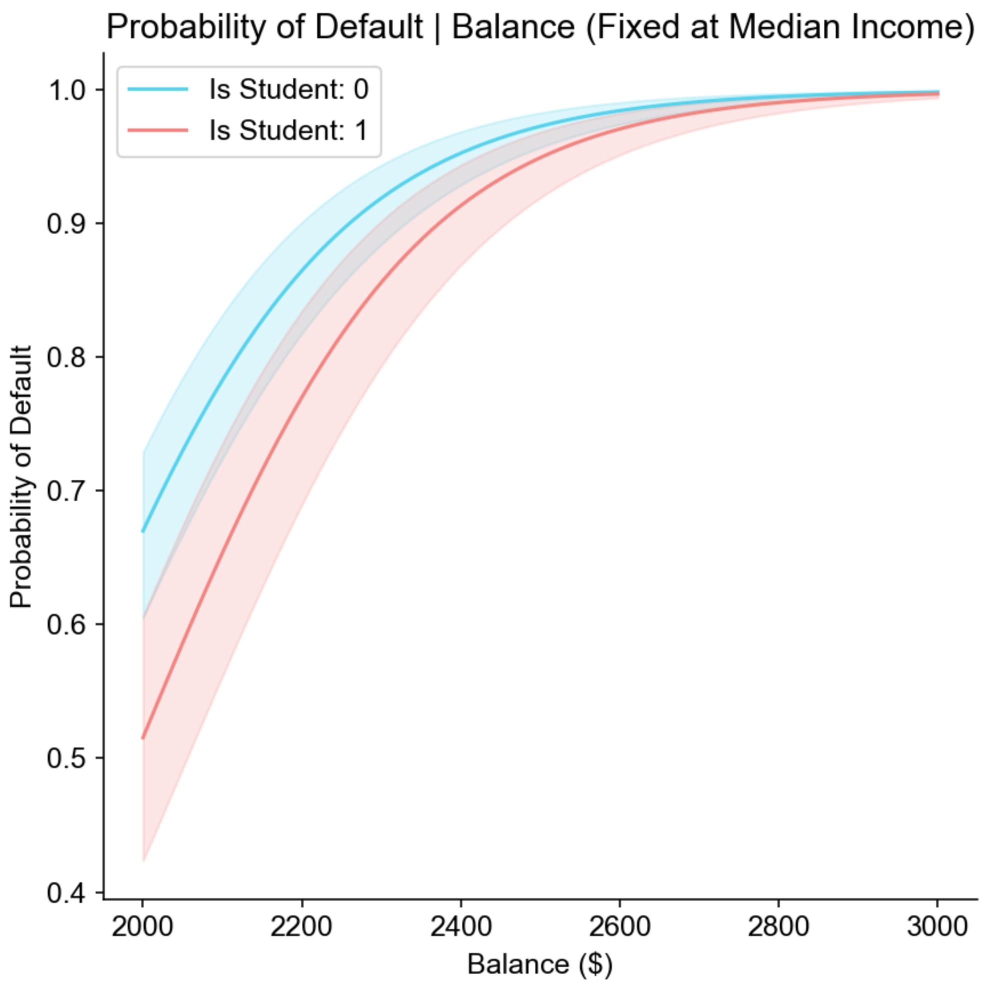
Logistic Regression Inference - Example 2

We can also perform inference about mean response (default) with categorical features by comparing the confidence intervals of several groups

In Introduction to Statistical Learning with Python, an example was provided for credit default dataset

Using the same model as estimated in the book, we provide the estimation result for balance and income for students and non-students

The coefficient of student is negative meaning that the default rate, keeping balance and income fixed, is lower for the student group, what we see in the diagram at a balance range between $2000 and $3000



Model Fit Summaries

Fisher Scoring Logistic Regression Fit

Total Fisher Scoring Iterations: 6
Log Likelihood: -111.6295
Beta 0 = intercept (bias): True

Example 1
(Applied Logistic Regression)

Fisher Scoring Logistic Regression Summary

Parameter	Estimate	Std. Error	Wald Statistic	P-value	Lower CI	Upper CI
intercept (bias)	0.8058	0.8452	0.9534	0.3404	-0.8507	2.4622
LWT	-0.0152	0.0064	-2.3641	0.0181	-0.0278	-0.0026
RACE_2	1.0811	0.4881	2.2151	0.0268	0.1245	2.0376
RACE_3	0.4806	0.3567	1.3475	0.1778	-0.2185	1.1797

Fisher Scoring Logistic Regression Fit

Total Fisher Scoring Iterations: 10
Log Likelihood: -785.7724
Beta 0 = intercept (bias): True

Example 2
(Intro to Statistical Learning)

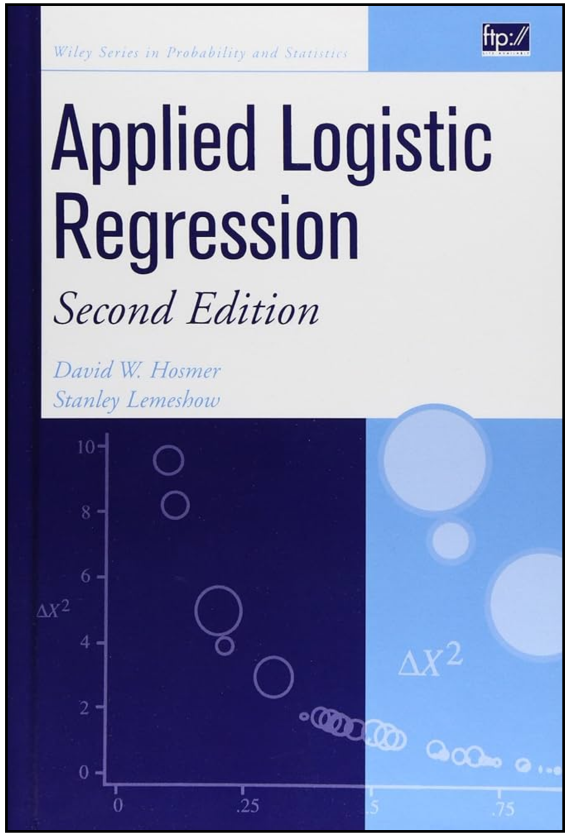
Fisher Scoring Logistic Regression Summary

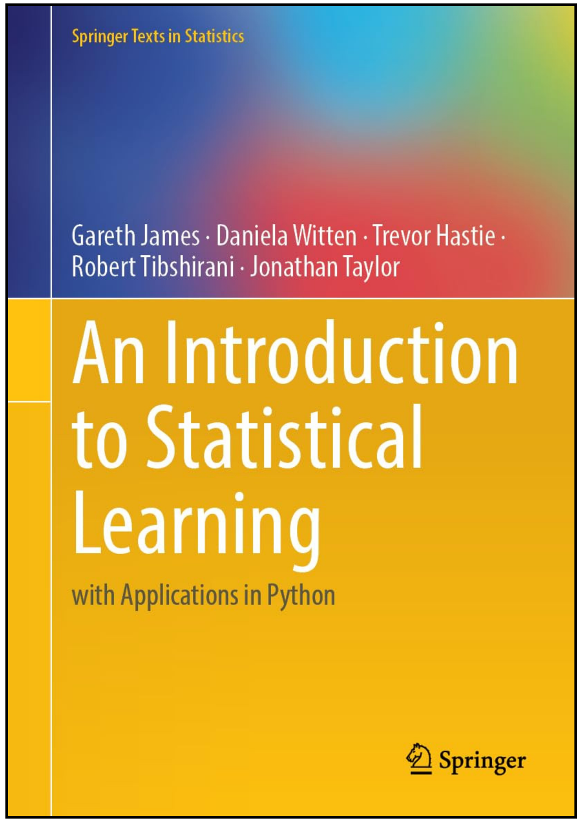
Parameter	Estimate	Std. Error	Wald Statistic	P-value	Lower CI	Upper CI
intercept (bias)	-10.8690	0.4923	-22.0793	0.0000	-11.8339	-9.9042
balance	0.0057	0.0002	24.7365	0.0000	0.0053	0.0062
income	0.0030	0.0082	0.3698	0.7115	-0.0130	0.0191
is_student	-0.6468	0.2363	-2.7376	0.0062	-1.1098	-0.1837

Reproducible Example

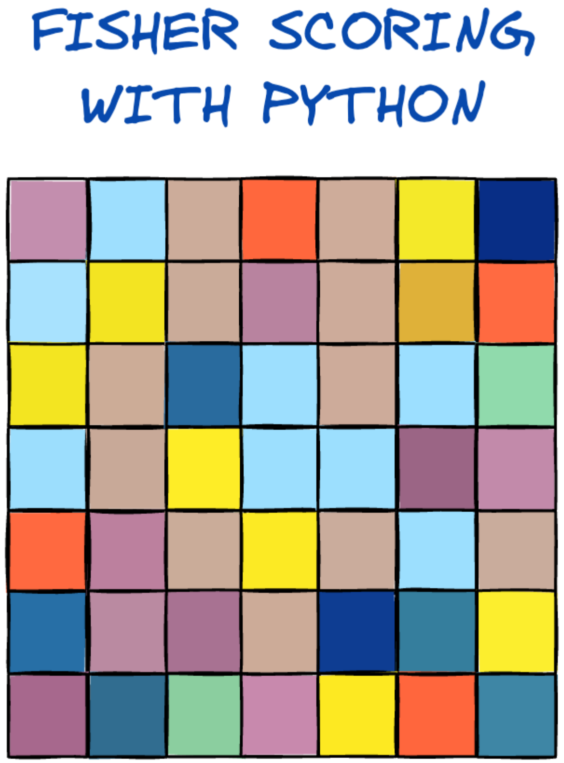
```python
from ISLP import load_data
from sklearn.model_selection import train_test_split
from fisher_scoring import FisherScoringLogisticRegression

Default = load_data('Default')

features = ['balance', 'income']
label = 'default'

X, y = Default[features], Default[label].map({'No': 0, 'Yes': 1})
X['is_student'] = Default['student'].map({'No': 0, 'Yes': 1})

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=0
)

# Fit model
model = FisherScoringLogisticRegression()
model.fit(X_train, y_train)

# Make predictions
probas = model.predict_proba(X_test)[:, 1]
intervals = model.predict_ci(X_test)
ci_lower, ci_upper = intervals[:, 0], intervals[:, 1]
```

Books


Wiley Series in Probability and Statistics
ftp://
Applied Logistic Regression
Second Edition
David W. Hosmer
Stanley Lemeshow
ΔX^2
ΔX^2


Springer Texts in Statistics
Gareth James · Daniela Witten · Trevor Hastie · Robert Tibshirani · Jonathan Taylor
An Introduction to Statistical Learning
with Applications in Python
Springer

Fisher Scoring with Python



github.com/xrisklab/fisher-scoring

Install with pip:

```
pip install fisher-scoring
```

Notebook with examples: